

Deep Learning for Medical Image Processing: Improving Chest X-ray Diagnostic Classification with Transfer Learning and Explainable Deep Learning

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Abstract

Chest X-ray imaging is one of the most widely used diagnostic tools for detecting thoracic abnormalities, yet manual interpretation can be time-consuming and subject to inter-observer variability. Deep learning has shown strong potential for assisting radiographic diagnosis, but model performance often depends on the availability of labeled data and model decisions are frequently difficult to interpret. This thesis investigates how transfer learning and explainable deep learning affect chest X-ray diagnostic classification in a focused and reproducible experimental setting. The study uses the NIH ChestX-ray14 dataset and formulates the task as binary classification, where images labeled “No Finding” are treated as normal and images with any pathology label are treated as abnormal. Two convolutional neural network architectures, ResNet50 and DenseNet121, are evaluated under four controlled experimental configurations: training from scratch and transfer learning for each architecture. Model performance is assessed using accuracy, precision, recall, F1-score, area under the ROC curve, and confusion-matrix-based error analysis. To improve interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) is applied to representative predictions in order to visualize the image regions that most strongly influence classification decisions. The final results show that DenseNet121 trained from scratch achieved the highest test AUC of 0.7457, while ResNet50 with transfer learning provided the best overall balance of accuracy, recall, and F1-score, reaching a test accuracy of 0.6955 and F1-score of 0.6547. These results indicate that transfer learning improved ResNet50 substantially, but its benefit was not uniform across architectures. The Grad-CAM analysis further showed

that the selected best-balance model generally focused on anatomically plausible thoracic regions while still exhibiting interpretable failure cases. Overall, the results suggest that controlled comparisons of training strategy and explainability can provide useful insights for capstone-scale medical image classification research.

Keywords: chest X-ray classification; transfer learning; deep learning; Grad-CAM; medical image analysis.

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Contents

Abstract	iii
Acknowledgements	v
List of Figures	ix
List of Tables	x
Definition of Acronyms	xi
Definition of Notations	xii
1 Introduction	1
1.1 Background	1
1.2 Motivation	2
1.3 Problem Statement	2
1.4 Research Objectives	3
1.5 Scope and Limitations	3
1.6 Research Questions	4
1.7 Thesis Structure	4
2 Literature Review	5
2.1 Deep Learning for Chest X-ray Classification	5
2.2 ResNet and DenseNet in Medical Imaging	7

2.3	Transfer Learning in Medical Image Analysis	8
2.4	Explainable Artificial Intelligence for Chest X-ray Models	10
2.5	Research Gap	11
2.6	Summary	12
3	Methodology	13
3.1	Research Design	13
3.2	System Workflow	14
3.3	Dataset Description	14
3.4	Data Preprocessing	15
3.5	Model Architectures	16
3.6	Training Strategies	17
3.7	Evaluation Metrics	18
3.8	Explainability with Grad-CAM	20
3.9	Implementation Environment	20
4	Experiments and Results	22
4.1	Experimental Setup	22
4.2	Baseline Results	23
4.3	Transfer Learning Results	24
4.4	Comparative Analysis	25
4.4.1	Comparison with Previous Literature	26
4.5	Training Curves and Error Analysis	27
4.6	Grad-CAM Visualization Results	32
5	Discussion, Conclusion, and Future Work	34
5.1	Interpretation of Results	34
5.2	Implications for Medical Image Classification	35
5.3	Limitations	36

5.4	Conclusion	37
5.5	Future Work	38
A	Supplementary Materials	40
A.1	Training Configuration Summary	40
A.2	Stored Experiment Artifacts	41
A.3	Additional Note on Grad-CAM Examples	42
	References	43

List of Figures

3.1	End-to-end workflow for the thesis pipeline.	14
4.1	Training and validation loss curves for the four full experiments.	28
4.2	Training and validation AUC curves for the four full experiments.	29
4.3	Test ROC curves for the four full experiments generated from per-sample prediction probabilities.	30
4.4	Final test confusion matrices for all four experimental configurations.	31
4.5	Grad-CAM for correct cases	32
4.6	Grad-CAM for error cases	32

List of Tables

4.1	Summary of experimental configurations	23
4.2	Baseline results for models trained from scratch	23
4.3	Results for models trained with transfer learning	24
4.4	Final comparison of all four experiments	25
A.1	Shared training configuration for the four main experiments.	41

Definition of Acronyms

Acronym	Definition
AUC	Area Under the Curve
CNN	Convolutional Neural Network
DL	Deep Learning
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
NIH	National Institutes of Health
ROC	Receiver Operating Characteristic
XAI	Explainable Artificial Intelligence

Definition of Notations

Symbol	Meaning
x	Input chest X-ray image
y	Ground-truth class label
$\hat{y}$	Predicted class label
$p(y = 1 \mid x)$	Predicted probability of the abnormal class
$\mathcal{L}$	Loss function
θ	Learnable model parameters

Chapter 1

Introduction

This thesis studies whether transfer learning and explainable deep learning can improve the performance and interpretability of chest X-ray diagnostic classification. The project is deliberately scoped as a binary classification problem on the NIH ChestXray14 dataset, where images labeled “No Finding” are treated as normal and all other labeled cases are treated as abnormal. Two convolutional neural network models, ResNet50 and DenseNet121, are compared under both training-from-scratch and transfer-learning settings, and Grad-CAM is used to visualize the basis of model predictions. The central practical question is not only whether a model can achieve acceptable aggregate metrics, but also whether it can reduce medically important failure cases such as missed abnormal radiographs.

1.1 Background

Chest X-ray imaging is one of the most widely used medical imaging modalities because it is inexpensive, fast, and routinely available in clinical settings. It plays an important role in screening and monitoring thoracic diseases, but manual interpretation can be time-consuming and subject to observer variability. As the number of medical images continues to increase, computer-aided diagnostic systems based on deep learning have become an

important research direction. In this context, errors are not equally serious. False positive predictions increase unnecessary follow-up and reduce trust in the system, but false negative predictions are more clinically concerning because an abnormal case may be missed and therefore not receive timely review or further diagnostic attention.

1.2 Motivation

Recent convolutional neural network architectures have shown strong performance in general image recognition and medical image analysis. However, chest X-ray classification still faces practical challenges such as label noise, class imbalance, and limited interpretability. In addition, performance gains reported in the literature often depend on transfer learning, yet the benefit of transfer learning should still be validated in a controlled experimental setting for the selected dataset and task. This motivates a focused comparison between models trained from scratch and models initialized with pretrained weights. It also motivates the use of an explainability method, because a model that appears numerically strong may still be unsuitable for screening if it fails to respond to subtle abnormal findings or attends to irrelevant image regions.

1.3 Problem Statement

The main problem addressed in this thesis is how to build a chest X-ray classification pipeline that is both effective and interpretable. Specifically, this work asks whether transfer learning can improve binary normal-versus-abnormal classification on NIH ChestXray14 and whether Grad-CAM can provide meaningful insight into the prediction behavior of the model. The problem is significant because diagnostic support systems must not only achieve strong quantitative performance but also provide evidence that their decisions are based on medically plausible image regions.

The research gap addressed by this thesis is intentionally narrow. Much of the recent chest X-ray literature emphasizes large multi-label benchmarks, newer architectures, or broad claims about transfer learning, but fewer compact studies isolate a controlled binary normal-versus-abnormal task while directly comparing scratch training, transfer learning, and Grad-CAM-based interpretation across two standard CNN baselines. This thesis addresses that gap through a reproducible capstone-scale comparison rather than through a new model proposal.

1.4 Research Objectives

The objectives of this thesis are as follows:

- formulate a clear binary chest X-ray classification task using NIH ChestXray14;
- implement two baseline architectures, ResNet50 and DenseNet121;
- compare training from scratch against transfer learning for both architectures;
- evaluate the models using standard classification metrics and error analysis;
- apply Grad-CAM to visualize representative successful and failed predictions.

1.5 Scope and Limitations

This study is limited to a single public dataset and a binary classification setting. It does not include lesion localization, segmentation, object detection, federated learning, or clinical deployment. The models are evaluated as research prototypes rather than as medical devices. Interpretability is examined through Grad-CAM, which is useful for qualitative analysis but does not provide a complete explanation of model reasoning.

1.6 Research Questions

This thesis is organized around the following research questions:

- How do ResNet50 and DenseNet121 perform on binary chest X-ray classification when trained from scratch?
- Does transfer learning improve the performance of these models on NIH ChestXray14?
- What types of prediction errors remain after training, and can Grad-CAM help explain those errors?

1.7 Thesis Structure

Chapter 2 reviews prior work on chest X-ray classification, transfer learning in medical imaging, and explainable artificial intelligence. Chapter 3 describes the dataset, preprocessing pipeline, model architectures, training settings, workflow, evaluation metrics, and Grad-CAM method. Chapter 4 presents the experimental setup, quantitative results, training curves, ROC analysis, confusion matrices, and Grad-CAM findings. Chapter 5 integrates the discussion, conclusion, and future-work directions into a single final chapter.

Chapter 2

Literature Review

This chapter reviews the main research areas that support this thesis: deep learning for chest X-ray analysis, transfer learning in medical image classification, and explainable artificial intelligence for model interpretation. The purpose of the review is not only to summarize prior work, but also to show why the present study is necessary and how its scope is justified. Since this capstone focuses on binary chest X-ray classification using ResNet50, DenseNet121, transfer learning, and Grad-CAM, the literature review is organized around those same themes.

2.1 Deep Learning for Chest X-ray Classification

Chest X-ray classification has become one of the most important applications of deep learning in medical imaging because chest radiography is widely available, relatively inexpensive, and frequently used in routine diagnosis. Unlike some imaging modalities that require expensive acquisition or complex annotation pipelines, chest X-ray datasets can be collected at scale, making them suitable for data-driven classification research. A major milestone in this area was the release of the ChestX-ray8 dataset, later expanded into ChestX-ray14, which provided a hospital-scale benchmark for thoracic disease classification and weakly supervised

localization¹. This dataset played an important role in making chest X-ray classification a standard benchmark problem in medical image analysis.

After the release of large public datasets, many studies applied convolutional neural networks to classify thoracic abnormalities from frontal chest radiographs. These studies differ in task formulation, with some focusing on binary classification and others using multi-class or multi-label settings. In practice, multi-label chest X-ray classification is more clinically realistic because one image may contain multiple findings. However, binary classification remains attractive for focused research because it simplifies the learning objective, reduces ambiguity in label interpretation, and makes controlled model comparison easier. For this reason, binary normal-versus-abnormal classification is a reasonable choice for an undergraduate capstone, especially when the goal is to compare model architectures and training strategies rather than to solve every aspect of the clinical problem at once.

One of the best-known chest X-ray studies is CheXNet, which used a DenseNet121-based model for pneumonia detection and demonstrated that deep neural networks could achieve strong predictive performance on a chest X-ray benchmark⁵. Later work such as CheXpert extended the discussion by focusing on uncertainty-aware labeling and more systematic benchmark design⁶. Comparative work by Baltruschat et al. further showed that model architecture choice, additional patient information, and training configuration can all affect chest X-ray classification performance on large public datasets⁷. Together, these studies show that chest X-ray classification is both practically important and technically challenging. They also reveal recurring issues, including label noise, dataset bias, class imbalance, and the gap between benchmark accuracy and trustworthy clinical use.

More recent work shows that the field is moving beyond standard CNN benchmarking toward broader evaluation settings, transformer-based models, and workflow-oriented validation. For example, recent studies have explored transformer-based multi-disease diagnosis¹⁵, anatomy-aware weakly supervised progression classification¹⁶, and few-shot augmentation

strategies designed to improve learning under limited supervision¹⁷. Other work has shifted attention from benchmark accuracy alone to the practical interaction between clinicians and AI systems, including comparisons between radiologists working with and without AI assistance¹⁸. In parallel, the CXR-LT 2024 challenge has reinforced current interest in long-tailed, multi-label, and zero-shot chest radiograph evaluation²⁵. These recent developments help position the present thesis within an active research area while also justifying the choice of a simpler and more controlled binary formulation.

2.2 ResNet and DenseNet in Medical Imaging

ResNet and DenseNet are among the most influential convolutional neural network architectures used in image classification. ResNet introduced residual learning, which allows deep networks to be optimized more effectively by learning residual mappings rather than attempting to fit every layer from scratch³. This design helped address optimization difficulties in very deep networks and made residual architectures a standard choice across computer vision tasks. Because of its strong performance and wide adoption, ResNet is often used as a baseline in medical imaging studies.

DenseNet extended this line of development by connecting each layer to all subsequent layers within a dense block, which improves feature reuse and strengthens gradient flow during training⁴. Dense connectivity can be particularly useful when data are limited, because later layers have direct access to earlier representations without requiring repeated relearning of similar low-level features. This property has made DenseNet especially popular in medical image classification, including chest X-ray analysis, where datasets can be large in raw image count but still limited in annotation quality and disease balance.

In chest X-ray research, both ResNet and DenseNet are common because they offer a strong tradeoff between performance, implementation simplicity, and computational feasibility. DenseNet121 became especially prominent through CheXNet⁵, while ResNet variants

remain widely used as reference baselines in medical imaging literature. For this thesis, selecting ResNet50 and DenseNet121 is appropriate for three reasons. First, both are mature and well-understood architectures. Second, both have pretrained ImageNet weights available, making transfer-learning experiments straightforward. Third, the comparison between them is academically meaningful while remaining realistic for a capstone project with limited time and computational resources.

2.3 Transfer Learning in Medical Image Analysis

Transfer learning is widely used in medical image analysis because the amount of labeled medical data is often much smaller than the amount of labeled natural-image data used to train large vision models. A common strategy is to initialize a model with ImageNet-pretrained weights and then fine-tune it on a target medical dataset. This approach can improve convergence speed, stabilize training, and sometimes improve final predictive performance, especially when annotations are limited or class imbalance is severe. Surveys of deep learning in medical image analysis consistently identify transfer learning as one of the most practical techniques for adapting general computer vision models to specialized clinical tasks².

Prior studies have shown that the benefits of transfer learning are real but not uniform. Pan and Yang provided one of the foundational surveys of transfer learning and clarified why transferring knowledge across domains can reduce the demand for expensive labeled data⁸. Tajbakhsh et al. compared full training and transfer-learning strategies across multiple medical imaging applications and found that transfer learning can be highly effective, particularly when labeled training data are scarce⁹. Other studies in radiology and computer-aided detection have also used pretrained CNN backbones to compensate for limited domain-specific supervision¹⁰. At the same time, the magnitude of transfer-learning gains depends on factors such as the similarity between source and target domains, the depth of fine-tuning, the size

of the target dataset, and the quality of labels.

This issue is especially relevant for the present thesis. Although NIH ChestXray14 contains many images, the labels are derived from report mining rather than direct expert annotation for every task, which introduces uncertainty and noise. In addition, some prior studies have explored whether high-capacity models can be trained effectively from scratch by exploiting label dependencies rather than relying entirely on pretraining¹¹. Therefore, comparing scratch training with transfer learning is not only a standard engineering choice, but also a meaningful research question. In this capstone, transfer learning is treated as a central variable rather than as an assumed default, because the goal is to measure whether pretrained representations provide a consistent advantage for the selected binary classification task.

Recent work has made this question even more relevant. A first trend is the growing use of self-supervised learning to reduce dependence on expensive annotations in chest X-ray imaging^{19,22}. A second trend is the evaluation of domain-specific pretraining against standard ImageNet initialization, with recent studies showing that chest-radiograph-specific representations do not always behave the same way as generic natural-image pretraining²³. Other work has explored more specialized transfer variants, including double-transfer learning for chest X-ray mass classification²⁶, while broader foundation-model studies have emphasized global-local feature integration and large-scale representation learning²⁷. Taken together, these developments reinforce the need to test whether transfer learning still provides a measurable benefit under a focused binary chest X-ray pipeline rather than assuming that the answer is always positive.

2.4 Explainable Artificial Intelligence for Chest X-ray Models

High predictive performance alone is insufficient for safety-critical applications such as medical diagnosis. In healthcare settings, clinicians and researchers need at least some visibility into why a model makes a decision, especially when a prediction may influence follow-up examination or treatment. As a result, explainable artificial intelligence has become an important companion to medical image classification research. Instead of treating the neural network as a complete black box, explainability methods attempt to identify which image regions or features most strongly influence a prediction.

Among these methods, Grad-CAM is one of the most widely used because it is practical, architecture-friendly, and relatively easy to interpret¹². Grad-CAM computes class-specific gradient information and projects it back onto feature maps in order to highlight image regions associated with the predicted class. In chest X-ray classification, this makes it possible to inspect whether the model is focusing on plausible thoracic regions or on irrelevant artifacts, borders, or text markers. Even though Grad-CAM is not a complete explanation of model reasoning, it provides a useful qualitative tool for detecting suspicious behavior and supporting error analysis.

Chest X-ray studies that include visual explanation often use saliency-based methods to demonstrate that a model is not merely fitting spurious patterns. However, explainability claims in the literature should still be treated carefully. A visually appealing heatmap does not guarantee causal understanding, and interpretation remains partly subjective. Recent surveys on explainable AI in healthcare and medical imaging also emphasize that transparency tools should be treated as supportive evidence rather than as complete validation of model trustworthiness^{13,14}. For this thesis, Grad-CAM is an appropriate choice because it adds an interpretable analysis layer without greatly increasing implementation complex-

ity. It fits the capstone scope well by supporting a focused examination of both correct predictions and failure cases.

This point is also reflected in recent chest-radiology studies. Some recent work has focused on improving explanation quality within weakly supervised localization settings, for example through Ensemble-CAM for thoracic disease localization²⁰. Other work has examined explainability from a human-centered perspective by studying how different XAI techniques are perceived in chest radiology workflows²¹. Recent explainable transformer-based chest X-ray models further show that modern architectures are now being designed together with explicit interpretation mechanisms rather than only adding visualization after training²⁴. At a broader level, recent systematic reviews continue to argue that explainability in medical imaging should be assessed not only by visual plausibility, but also by usability and methodological rigor²⁸. These studies support the decision to include Grad-CAM in the thesis as a practical interpretability component, while still acknowledging its limitations.

2.5 Research Gap

The literature reviewed above shows that chest X-ray classification is already a mature research area, but it also reveals several opportunities for a focused study. First, many papers emphasize headline performance on large benchmark tasks, while fewer present a clean and controlled comparison between widely used baseline architectures under both scratch-training and transfer-learning settings. Second, chest X-ray papers often prioritize predictive accuracy, whereas interpretability and error analysis are discussed more briefly or only through a few visual examples. Third, large multi-label formulations can make it difficult to isolate the effect of architecture and training strategy.

This thesis addresses a narrower but useful gap by conducting a compact comparative study centered on one public dataset, one binary task, two standard CNN architectures, and one explainability method. Rather than proposing a new architecture, the study aims

to answer a practical question: for binary normal-versus-abnormal chest X-ray classification on NIH ChestXray14, how much do architecture choice, transfer learning, and Grad-CAM-based interpretation contribute to a reliable and understandable pipeline? This narrower framing is suitable for a capstone because it favors clarity, reproducibility, and defensible analysis over unnecessary methodological sprawl.

2.6 Summary

The literature shows that chest X-ray classification is an important and active research area in medical image analysis. Public datasets such as ChestXray14 have enabled large-scale benchmarking, while architectures such as ResNet and DenseNet have become standard choices for visual recognition tasks. Prior work also shows that transfer learning is often beneficial in medical imaging, although its impact depends on the exact dataset and training setup. At the same time, explainability remains essential because strong predictive performance alone is insufficient for trustworthy medical AI.

Based on this review, the methodological choices in this thesis are well justified. ResNet50 and DenseNet121 provide meaningful and established baselines, transfer learning is a central experimental variable, and Grad-CAM offers a practical way to inspect model behavior. The next chapter therefore describes how these components are implemented and evaluated on NIH ChestXray14 for binary chest X-ray classification.

Chapter 3

Methodology

3.1 Research Design

This thesis adopts a comparative experimental design to evaluate chest X-ray diagnostic classification under a controlled setting. The central goal is to determine whether transfer learning improves binary chest X-ray classification performance and whether explainability methods can make model behavior easier to interpret. To ensure that the comparison is fair, all experiments use the same dataset, label definition, preprocessing pipeline, split strategy, and evaluation metrics. The only differences between the core experiments are the network architecture and the training strategy.

Four main experiments are defined in this study. The first and second experiments use ResNet50, while the third and fourth experiments use DenseNet121. For each architecture, one model is trained from scratch and one model is trained with ImageNet-pretrained weights. This yields the following comparison structure:

- ResNet50 trained from scratch;
- ResNet50 with transfer learning;

- DenseNet121 trained from scratch;
- DenseNet121 with transfer learning.

This design makes it possible to analyze two questions directly: whether one architecture is more suitable for the selected task, and whether transfer learning provides a consistent performance advantage across both architectures.

3.2 System Workflow

Figure 3.1 summarizes the end-to-end workflow used in this thesis. The pipeline begins with NIH ChestXray14 image-label pairs, applies binary label mapping and preprocessing, creates a fixed train-validation-test split, trains the four controlled model configurations, and then evaluates the final models using both quantitative metrics and Grad-CAM-based qualitative analysis. This workflow is intentionally simple so that the full comparison remains reproducible and easy to audit.

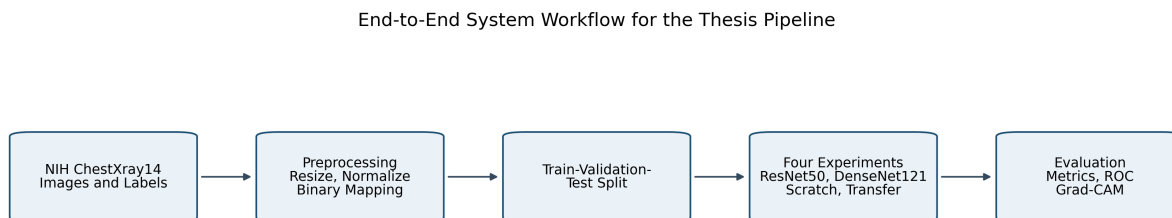


Figure 3.1: End-to-end workflow for the thesis pipeline.

3.3 Dataset Description

The experiments are based on the NIH ChestXray14 dataset, which is one of the most widely used public benchmarks for chest radiograph analysis¹. The dataset contains a large collection of frontal chest X-ray images with labels mined from radiology reports. Because the original dataset is multi-label and includes several thoracic disease findings, it is useful

for many different classification settings. However, for the purpose of this capstone, the task is simplified to binary classification so that the effect of architecture choice, transfer learning, and explainability can be studied in a more controlled way.

In this thesis, all images labeled “No Finding” are assigned to the *Normal* class. Images containing one or more pathology labels are assigned to the *Abnormal* class. This mapping converts the original multi-label problem into a binary normal-versus-abnormal decision problem. The binary setting is a practical choice for three reasons. First, it reduces ambiguity in the learning objective. Second, it allows clearer interpretation of evaluation metrics such as sensitivity and specificity. Third, it is appropriate for a focused capstone project where the main goal is comparative analysis rather than full clinical deployment.

This label mapping should be interpreted with caution. The NIH ChestXray14 labels are mined from radiology reports rather than manually annotated for this exact binary task, so label noise is expected. In addition, “No Finding” does not necessarily mean that the patient is completely healthy; it only means that no corresponding abnormality was extracted from the associated report. Therefore, the *Normal* and *Abnormal* classes used in this thesis are operational research labels for a controlled experiment, not definitive clinical ground truth.

3.4 Data Preprocessing

The data preprocessing pipeline is designed to produce reproducible model inputs while preserving the overall structure of the chest radiographs. First, metadata are inspected to identify the image path, patient identifier if available, and original disease labels. After the binary label mapping is applied, the dataset is divided into training, validation, and test subsets using a target split ratio of 70%, 15%, and 15%, respectively. If patient identifiers are available, patient-level splitting is used to reduce the risk of data leakage between training and evaluation sets.

All images are resized to 224×224 pixels so that they can be processed consistently by both ResNet50 and DenseNet121. Since chest X-ray images are grayscale but the selected pretrained models are designed for three-channel input, the image is converted into a model-compatible tensor format while preserving the underlying grayscale information. Pixel values are normalized after resizing so that training remains stable across different model configurations.

To reduce overfitting and improve robustness, training-time augmentation is applied to the training subset only. The final augmentation strategy includes resizing, normalization, and a modest set of geometric or intensity-preserving transformations that do not distort the clinical meaning of the radiograph. The validation and test sets use deterministic preprocessing only, so that the evaluation remains stable and comparable across runs. Every preprocessing step is documented so that the full experimental pipeline can be reproduced later.

3.5 Model Architectures

Two convolutional neural network architectures are used in this study: ResNet50³ and DenseNet121⁴. These models are selected because they are well established, widely used in medical image classification, and supported by pretrained ImageNet weights. They also provide a meaningful comparison between two influential design philosophies in convolutional neural networks.

ResNet50 is based on residual learning, where shortcut connections allow information to bypass one or more layers. This helps stabilize optimization in deep networks and makes residual models strong baselines across many vision tasks. In the context of this thesis, ResNet50 is included because it represents a mature and widely adopted architecture that can serve as a reliable benchmark.

DenseNet121 uses dense connectivity, in which each layer receives feature information from earlier layers in the same dense block. This design promotes feature reuse and improves gradient flow during training. DenseNet121 is especially relevant to chest X-ray analysis because it was used by CheXNet and has become one of the most recognizable baselines in this area⁵. In all experiments, the final classification layer of each network is adapted to output the binary classes required by the selected task.

3.6 Training Strategies

Two training strategies are compared for each model. In the first strategy, the network is trained from scratch using random parameter initialization. In the second strategy, the network is initialized with ImageNet-pretrained weights and then fine-tuned on the target chest X-ray dataset. This comparison is motivated by prior medical imaging work showing that transfer learning often improves convergence and performance when labeled data are limited or noisy^{9,10}.

The final training configuration used a batch size of 16, a learning rate of 10^{-3} , the Adam optimizer, and `BCEWithLogitsLoss` for binary classification. Each experiment was trained for up to 20 epochs, and the best checkpoint was selected based on validation AUC. An early stopping mechanism with patience set to 5 was used to reduce unnecessary computation and to limit overfitting. The same optimization procedure was applied consistently across the four core comparisons.

The same learning rate was used across all experiments to preserve comparability between architectures and training strategies. This design choice keeps the comparison focused on architecture and initialization rather than on separately tuned hyperparameters. However, for pretrained CNN fine-tuning, a lower learning rate can sometimes be more appropriate. Separate learning-rate tuning for scratch and transfer-learning settings is therefore treated as a direction for future work rather than as part of the controlled comparison in this thesis.

For transfer learning, the pretrained model was adapted by replacing the original final classification head with a binary output layer and then fine-tuned on the target dataset under the same experimental conditions used for the scratch-training baseline. Weighted loss was not introduced into the four main comparisons because the goal of the thesis was to preserve a controlled comparison focused on architecture choice and training strategy.

Algorithmic procedure. The practical training procedure used throughout the thesis can be summarized as follows:

1. Load the predefined training and validation splits and apply the corresponding transforms.
2. Initialize either ResNet50 or DenseNet121 with scratch weights or ImageNet-pretrained weights.
3. Replace the original classification head with a binary output layer.
4. Train the model for one epoch using mini-batch Adam optimization and binary cross-entropy loss.
5. Evaluate the model on the validation split and record validation AUC.
6. Save the checkpoint if validation AUC improves; otherwise increment the early-stopping counter.
7. Stop training when the patience limit is reached or the maximum number of epochs is completed.

3.7 Evaluation Metrics

Model performance is evaluated using a set of standard binary classification metrics. These include accuracy, precision, recall, F1-score, and the area under the ROC curve (AUC).

Accuracy measures overall correctness, while precision and recall provide complementary views of positive prediction quality and sensitivity to abnormal cases. F1-score is useful because it balances precision and recall, especially when class distributions are uneven. AUC is included because it measures ranking quality across decision thresholds and is frequently reported in chest X-ray studies^{1,6}.

In addition to these summary metrics, confusion matrices are used to analyze false positives and false negatives. Sensitivity and specificity are also reported because they are especially meaningful in a diagnostic setting. Together, these metrics allow the thesis to compare models not only by final predictive performance but also by error behavior and class-specific tradeoffs.

The binary classification loss used in the experiments is `BCEWithLogitsLoss`, which combines a sigmoid activation with binary cross-entropy:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \sigma(z_i) + (1 - y_i) \log (1 - \sigma(z_i))], \quad (3.1)$$

where z_i is the model logit for sample i , $\sigma(\cdot)$ is the sigmoid function, and $y_i \in \{0, 1\}$ is the binary class label.

The main evaluation metrics are computed from the standard confusion-matrix counts true positive (TP), true negative (TN), false positive (FP), and false negative (FN):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (3.2)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (3.3)$$

$$\text{Recall} = \text{Sensitivity} = \frac{TP}{TP + FN}, \quad (3.4)$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (3.5)$$

$$\text{Specificity} = \frac{TN}{TN + FP}. \quad (3.6)$$

The AUC is not tied to a single decision threshold; instead, it summarizes ranking quality across the full ROC curve by measuring how well abnormal cases are ranked above normal cases.

3.8 Explainability with Grad-CAM

Explainability is incorporated into the methodology through Grad-CAM¹². After the quantitative evaluation is completed, Grad-CAM is applied to representative images from the test set in order to visualize which image regions most strongly influence model predictions. This process is used for both correctly classified and misclassified cases. By analyzing correct normal predictions, correct abnormal predictions, false positives, and false negatives, the study can examine whether the model focuses on plausible thoracic regions or relies on irrelevant patterns.

The purpose of Grad-CAM in this thesis is not to claim perfect interpretability, but to provide qualitative evidence about model behavior. This is particularly important in medical imaging, where an apparently strong model may still behave unreliably if its predictions are driven by spurious visual cues. Grad-CAM therefore complements the numerical results by supporting a focused error analysis and a discussion of model trustworthiness.

3.9 Implementation Environment

The implementation environment for this study is Python with standard deep learning and data analysis libraries. The core framework is PyTorch, supported by torchvision for pre-trained model access and image transforms. Additional tools include NumPy and pandas for data handling, scikit-learn for evaluation metrics, and matplotlib for visualization. Model checkpoints, result tables, training logs, ROC curves, confusion matrices, and Grad-CAM

figures are stored in the project workspace so that results can be reused directly in later chapters of the thesis.

Overall, the methodology is designed to balance rigor and feasibility. The project does not attempt to solve every problem in chest X-ray analysis. Instead, it uses a narrow but defensible experimental design to answer a focused research question: whether standard convolutional architectures, combined with transfer learning and Grad-CAM, can provide an effective and interpretable framework for binary chest X-ray diagnostic classification.

Chapter 4

Experiments and Results

4.1 Experimental Setup

This chapter presents the experimental results for the four core model configurations defined in Chapter 3. The goal is to compare architecture choice and training strategy under a common evaluation setting. All experiments use the same binary classification task, the same preprocessing pipeline, and the same train-validation-test split. This ensures that any observed performance differences can be interpreted mainly in terms of the model architecture or the use of transfer learning rather than changes in data preparation.

The four experiments evaluated in this thesis are listed below:

- E1: ResNet50 trained from scratch;
- E2: ResNet50 with transfer learning;
- E3: DenseNet121 trained from scratch;
- E4: DenseNet121 with transfer learning.

The experiments reported in this chapter use the finalized NIH ChestXray14 binary split

introduced in Chapter 3, with 78,202 training images, 17,099 validation images, and 16,819 test images. All four experiments were trained with an image size of 224, batch size of 16, Adam optimizer, learning rate of 0.001, weight decay of 0.0001, and a maximum of 20 epochs, with early stopping applied when no validation improvement was observed for five consecutive epochs. The same loss function, preprocessing pipeline, and evaluation metrics were retained across runs to keep the comparison controlled and interpretable.

Table 4.1: Summary of experimental configurations

Experiment ID	Model	Training Strategy	Pretrained Weights
E1	ResNet50	From scratch	No
E2	ResNet50	Transfer learning	ImageNet
E3	DenseNet121	From scratch	No
E4	DenseNet121	Transfer learning	ImageNet

4.2 Baseline Results

This section reports the results of the two scratch-training experiments. These baselines are important because they establish how well each model performs without relying on pretrained visual features. They also provide a direct reference point for evaluating the later transfer-learning experiments.

The completed scratch runs show that DenseNet121 trained from scratch slightly outperformed ResNet50 trained from scratch across all major test metrics, including accuracy, recall, F1-score, and AUC. This is an important result because it demonstrates that pre-trained weights were not uniformly necessary for strong performance under the binary task formulation used in this thesis.

Table 4.2: Baseline results for models trained from scratch

Model	Accuracy	Precision	Recall	F1-score	AUC
ResNet50	0.6732	0.6748	0.5596	0.6118	0.7225
DenseNet121	0.6912	0.6913	0.5945	0.6393	0.7457

Among the two scratch baselines, DenseNet121 delivered the stronger result. It achieved a test AUC of 0.7457 compared with 0.7225 for ResNet50, while also producing higher accuracy and F1-score. This suggests that DenseNet121 was able to learn a more effective feature representation from the dataset even without external pretraining. At the same time, both scratch baselines remained clinically imperfect, particularly in recall, which means that a substantial fraction of abnormal cases would still be missed under the current threshold.

4.3 Transfer Learning Results

This section presents the results obtained when ResNet50 and DenseNet121 were initialized with ImageNet-pretrained weights. The purpose is to determine whether transfer learning improves performance relative to scratch training and whether the gains are consistent across both architectures.

Transfer learning was evaluated using the same test metrics for both architectures. In the completed experiments, both pretrained models achieved test AUC values above 0.74, showing that pretrained visual representations transferred effectively to this binary chest X-ray task. However, the two models produced noticeably different error profiles. ResNet50 delivered the strongest overall balance across accuracy, recall, F1-score, and AUC, whereas DenseNet121 achieved higher precision and specificity at the cost of substantially lower recall.

Table 4.3: Results for models trained with transfer learning

Model	Accuracy	Precision	Recall	F1-score	AUC
ResNet50 + transfer learning	0.6955	0.6847	0.6272	0.6547	0.7454
DenseNet121 + transfer learning	0.6711	0.7225	0.4634	0.5646	0.7421

The ResNet50 transfer-learning run completed all 20 epochs and achieved its best validation AUC at epoch 20. On the test set, it reached an AUC of 0.7454, accuracy of 0.6955, and F1-score of 0.6547. These values indicate that the model maintained a relatively balanced tradeoff between sensitivity and specificity. The DenseNet121 transfer-learning run

was resumed on a second machine from the checkpoint saved after epoch 7 and stopped early at epoch 17. Its best validation AUC was 0.7348 at epoch 12, and its final test AUC was 0.7421. Although the AUC difference between the two models is small, the DenseNet121 model produced a much lower recall of 0.4634, which indicates that more abnormal cases were missed under the current threshold.

At the same time, DenseNet121 achieved higher precision (0.7225) and specificity (0.8482) than ResNet50, suggesting that it was more conservative in assigning the abnormal label. This tradeoff is important in a medical screening context. A higher-recall model reduces the risk of missed abnormal cases, whereas a higher-specificity model reduces the number of false alarms. Under the binary formulation used in this study, ResNet50 appears to be the better balanced model, while DenseNet121 may be preferred only when the reduction of false positives is the dominant requirement.

4.4 Comparative Analysis

After presenting baseline and transfer-learning results separately, all four experiments can now be compared in one integrated analysis. This section is the core of the chapter because it connects the numerical results back to the thesis research questions. The goal is not only to identify the strongest model, but also to explain the tradeoffs between performance, efficiency, and interpretability.

Table 4.4: Final comparison of all four experiments

Experiment	Accuracy	Precision	Recall	F1-score	AUC	Sensitivity	Specificity
E1: ResNet50 scratch	0.6732	0.6748	0.5596	0.6118	0.7225	0.5596	0.7700
E2: ResNet50 transfer	0.6955	0.6847	0.6272	0.6547	0.7454	0.6272	0.7537
E3: DenseNet121 scratch	0.6912	0.6913	0.5945	0.6393	0.7457	0.5945	0.7736
E4: DenseNet121 transfer	0.6711	0.7225	0.4634	0.5646	0.7421	0.4634	0.8482

The four-way comparison reveals two important patterns. First, the best result depends on the evaluation priority. DenseNet121 trained from scratch achieved the highest overall

test AUC at 0.7457, narrowly exceeding ResNet50 with transfer learning at 0.7454. However, ResNet50 with transfer learning produced the highest accuracy, recall, and F1-score, which made it the most balanced model for the screening-oriented objective of detecting abnormal cases while avoiding an excessive precision-recall tradeoff.

Second, transfer learning did not affect the two architectures in the same way. For ResNet50, transfer learning improved every major headline metric relative to the scratch baseline, raising test accuracy from 0.6732 to 0.6955, recall from 0.5596 to 0.6272, F1-score from 0.6118 to 0.6547, and AUC from 0.7225 to 0.7454. For DenseNet121, the opposite pattern appeared: the scratch model slightly outperformed the transfer-learning model in accuracy, recall, F1-score, and AUC, while the transfer-learning version achieved notably higher precision and specificity. This suggests that ImageNet pretraining made DenseNet121 more conservative under the current threshold rather than uniformly stronger.

Taken together, the final results suggest that transfer learning is clearly beneficial for ResNet50, but not uniformly beneficial for DenseNet121 in this binary formulation. If the main goal is a balanced screening-oriented classifier, ResNet50 with transfer learning remains the most defensible choice because of its stronger recall and F1-score. If AUC alone is prioritized, DenseNet121 trained from scratch is marginally strongest. These tradeoffs justify the later discussion of threshold sensitivity, task formulation, and explainability.

4.4.1 Comparison with Previous Literature

The completed experiments are broadly consistent with prior literature in two ways. First, they support the observation that both ResNet and DenseNet remain strong baselines for chest radiograph classification^{1,7}. DenseNet121 remained highly competitive in this thesis and achieved the highest test AUC when trained from scratch, which is consistent with the strong reputation of DenseNet-based models in chest X-ray studies such as CheXNet⁵. Second, the ResNet50 results support earlier transfer-learning studies in medical imaging

showing that pretrained features can improve convergence and final classification quality when the downstream task is limited in scope or affected by label noise^{9,10}.

At the same time, the present findings also differ from the most generic transfer-learning expectation. In this thesis, transfer learning clearly improved ResNet50, but DenseNet121 trained from scratch slightly outperformed DenseNet121 with transfer learning in accuracy, recall, F1-score, and AUC. This suggests that the benefit of pretraining depends on the interaction between architecture, label formulation, and decision threshold rather than acting as an automatic improvement in every setting. The result remains defensible in light of more recent work showing that chest-radiograph transfer strategies can be architecture-sensitive and can interact with domain-specific pretraining in non-uniform ways^{19,22,23,27}. In the same way, the Grad-CAM findings are aligned with recent XAI literature: the heatmaps often highlight plausible thoracic regions, but they should still be interpreted as supportive evidence rather than as complete proof of correct clinical reasoning^{12,14,20,21,28}.

4.5 Training Curves and Error Analysis

This section reports the optimization and error-analysis figures used to interpret the completed runs. Together, the training curves, ROC curves, and confusion matrices clarify how the four experiments converged and where their classification tradeoffs emerged.

Figures 4.1 and 4.2 show that all four experiments converged in a stable way, but not with identical dynamics. ResNet50 with transfer learning improved steadily throughout training and continued to gain validation AUC until the final epoch, which is consistent with its strong balanced test performance. DenseNet121 trained from scratch also improved consistently and reached its best validation AUC at the end of training, which helps explain why it achieved the highest final test AUC. In contrast, DenseNet121 with transfer learning peaked earlier and then plateaued, which is consistent with its early-stopped training trajectory and lower final recall. The scratch ResNet50 baseline converged more slowly and remained the weakest

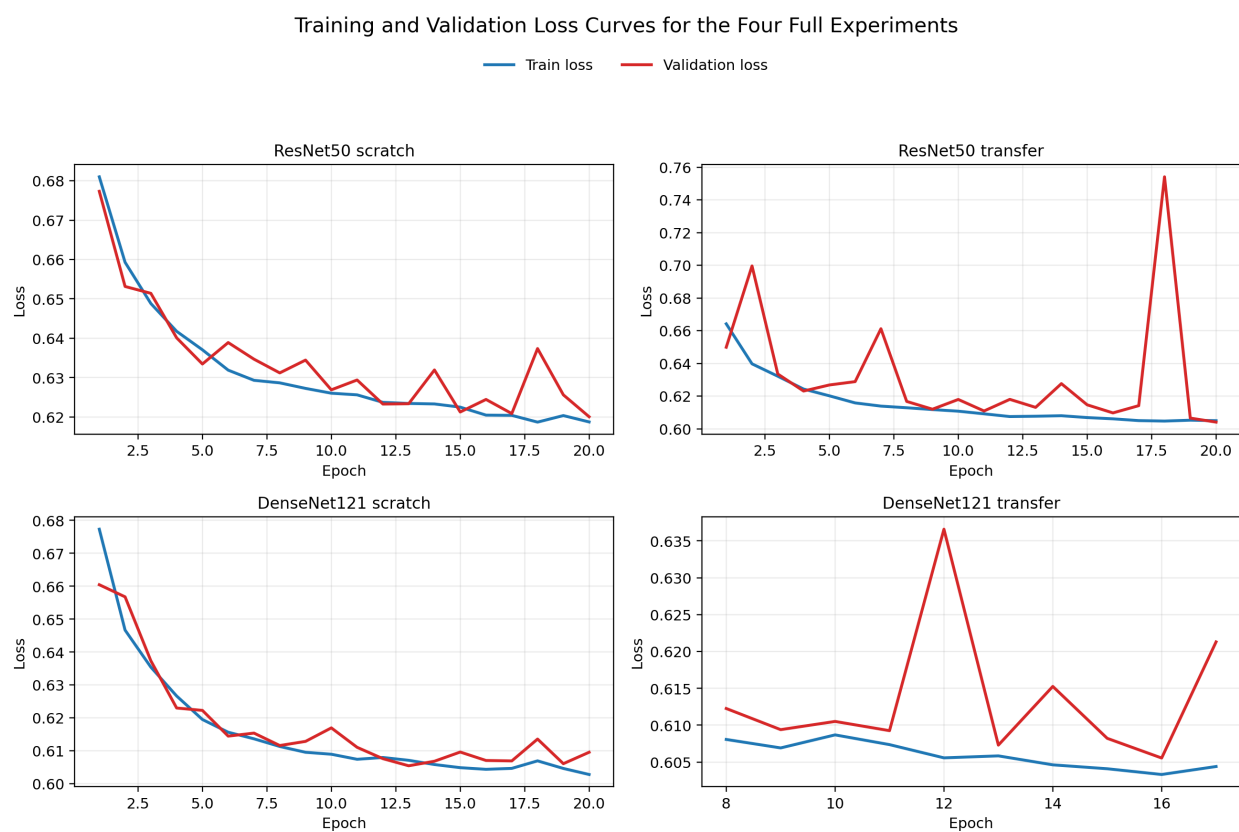


Figure 4.1: Training and validation loss curves for the four full experiments.

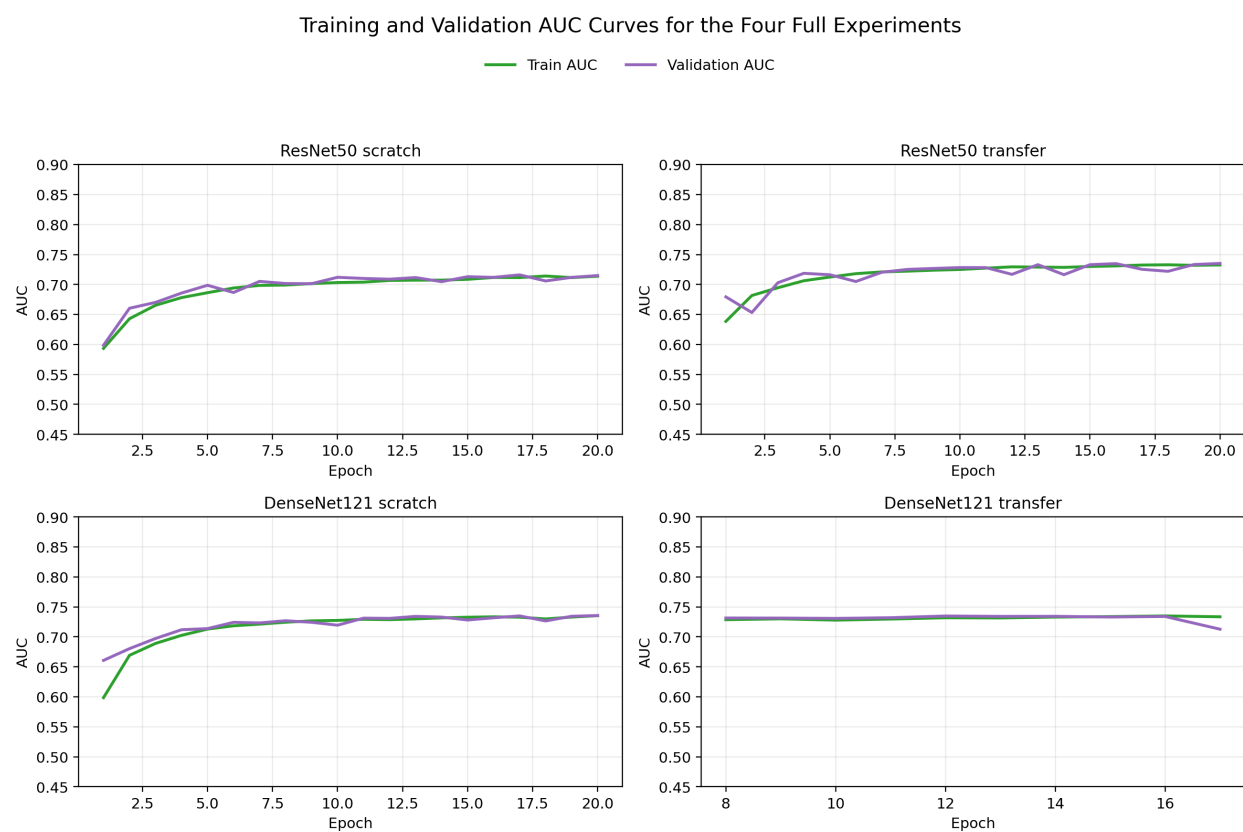


Figure 4.2: Training and validation AUC curves for the four full experiments.

of the four models across the main metrics.

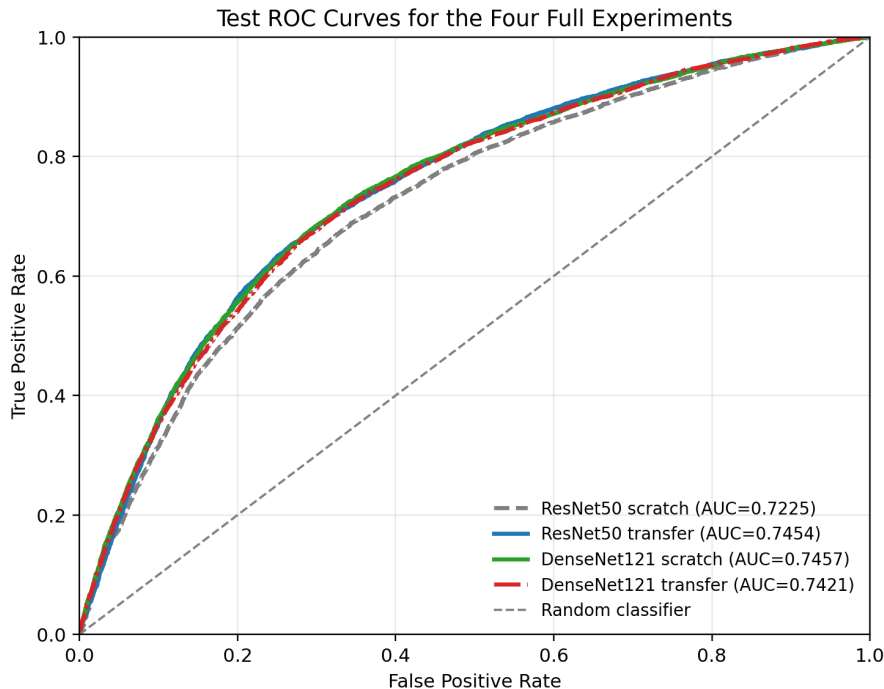


Figure 4.3: Test ROC curves for the four full experiments generated from per-sample prediction probabilities.

Figure 4.3 shows the ROC comparison for all four completed experiments using the per-sample prediction probabilities exported from the final test-set evaluations. This makes the ROC analysis consistent with the four-model design. DenseNet121 trained from scratch achieved the highest AUC (0.7457), ResNet50 with transfer learning was nearly identical (0.7454), DenseNet121 with transfer learning remained close (0.7421), and ResNet50 trained from scratch was lower (0.7225). The close AUC values among the strongest three models indicate similar ranking ability across thresholds, but the confusion matrices in Figure 4.4 show that their default-threshold error profiles differ substantially.

The confusion matrices in Figure 4.4 make the screening tradeoff especially clear. ResNet50 with transfer learning generated the largest number of correctly detected abnormal cases among the models with strong overall balance, which is why it remains the preferred screening-oriented configuration. DenseNet121 with transfer learning produced the fewest false pos-

Test Confusion Matrices Derived from the Final Experimental Results

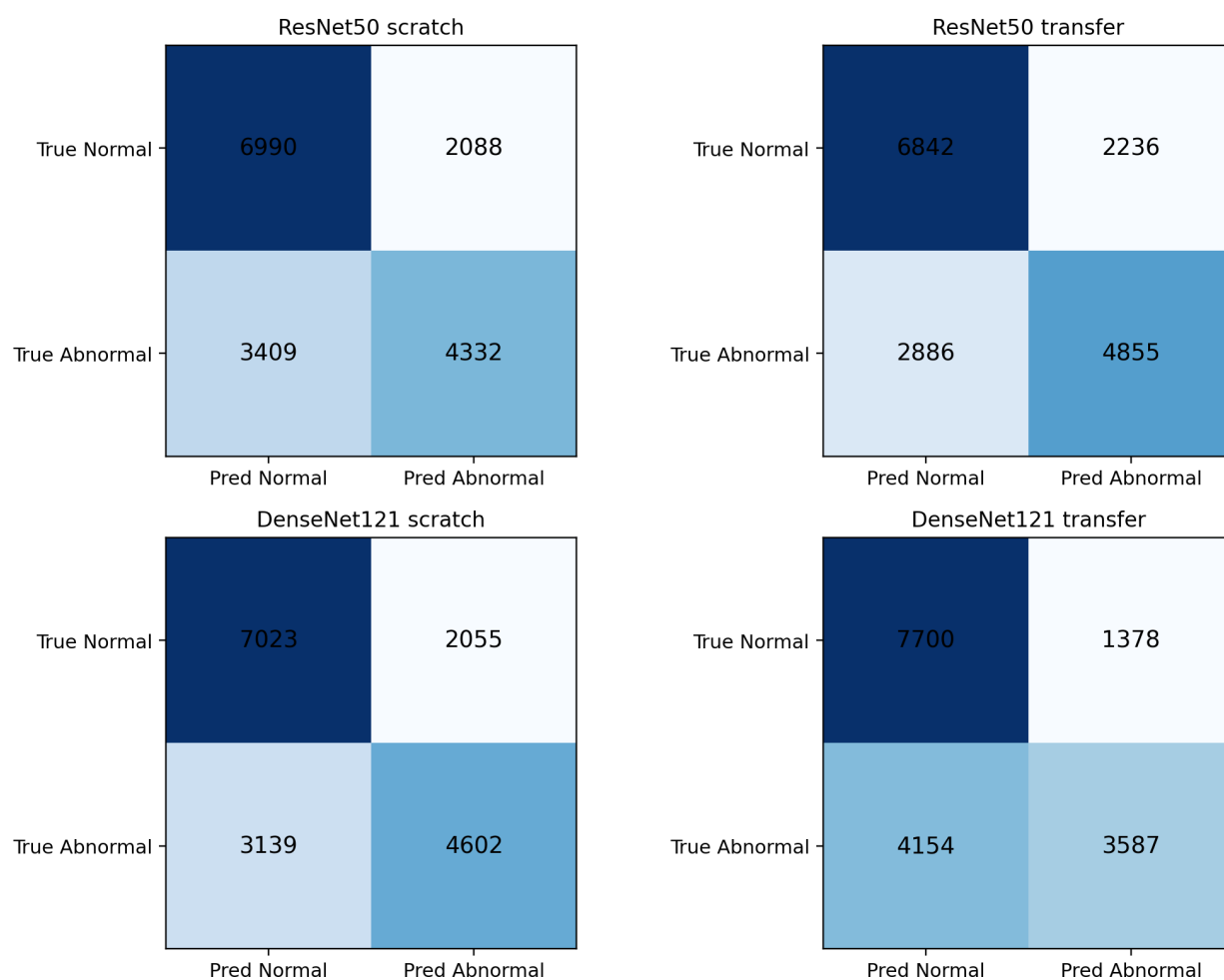


Figure 4.4: Final test confusion matrices for all four experimental configurations.

itives, but it also produced the most false negatives, which makes it less attractive when missed abnormal cases are treated as the more serious error. DenseNet121 trained from scratch occupied a more moderate position, combining strong AUC with better recall than its transfer-learning counterpart.

4.6 Grad-CAM Visualization Results

Grad-CAM analysis was conducted on the selected best-balance model, ResNet50 with transfer learning, in order to examine whether the model attended to plausible thoracic regions when making binary decisions. Four representative test cases were selected to cover correct normal prediction, correct abnormal prediction, false positive, and false negative. This set was chosen so that the visualization analysis would reflect both model strengths and model failure modes.



Figure 4.5: Grad-CAM examples for correctly classified normal and abnormal chest X-rays using the ResNet50 transfer-learning model.

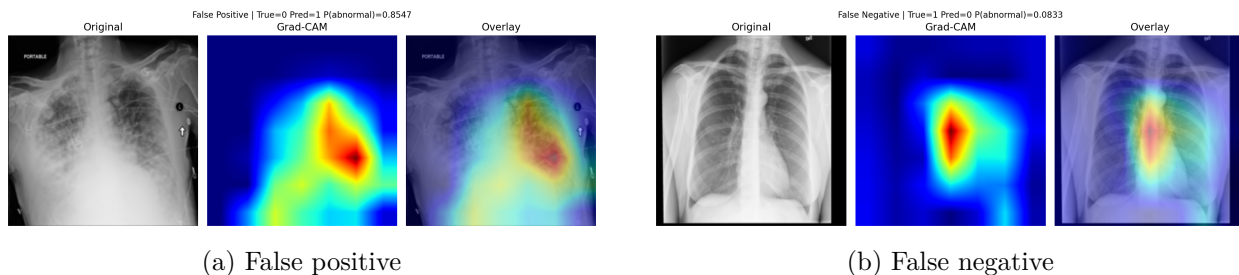


Figure 4.6: Grad-CAM examples for representative error cases produced by the ResNet50 transfer-learning model.

In the correct normal case shown in Figure 4.5a, the model focuses mainly on the central

thoracic region rather than on borders or obvious acquisition artifacts. This suggests that the classifier is relying on broad chest structure when making a normal decision. In the correct abnormal case shown in Figure 4.5b, the strongest activation is located in clinically relevant thoracic areas, with more emphasis placed on the right lung field. This behavior is consistent with the expectation that abnormal predictions should be driven by localized image evidence inside the chest region.

The error cases reveal a more mixed picture. In the false positive example in Figure 4.6a, the heatmap is relatively diffuse and weighted toward one hemithorax, which suggests that the model may sometimes overreact to intensity variation, positioning, or imaging artifacts in otherwise normal radiographs. In the false negative example in Figure 4.6b, the activation remains narrow and centrally concentrated, indicating that the model may fail to respond strongly enough when abnormal findings are subtle or do not dominate the overall appearance of the chest X-ray.

Overall, the Grad-CAM results provide partial support for the interpretability of the selected best-balance model. The model generally attends to anatomically plausible thoracic regions rather than to entirely irrelevant background areas, which is encouraging. At the same time, the error cases show that plausible attention maps do not guarantee correct prediction, and Grad-CAM should therefore be interpreted as supportive evidence rather than definitive clinical explanation.

Chapter 5

Discussion, Conclusion, and Future Work

5.1 Interpretation of Results

This final chapter interprets the experimental findings presented in Chapter 4 and connects them back to the research questions stated in Chapter 1. With all four planned experiments completed, the thesis can compare both architecture choice and training strategy within a single controlled pipeline. This makes it possible to discuss not only the value of pretrained representations, but also the situations in which scratch training may remain competitive.

Across the four experiments, ResNet50 with transfer learning achieved the best overall balance of predictive performance. It produced the highest test accuracy, recall, and F1-score, while DenseNet121 trained from scratch achieved the highest test AUC by a very small margin. DenseNet121 with transfer learning produced lower recall but the highest precision and specificity. This indicates that the architectures and training strategies reached different operating points under the same binary formulation. In practical terms, ResNet50 with transfer learning was better aligned with the screening-oriented objective of reducing missed abnormal cases, while DenseNet121 with transfer learning was stronger at avoiding unnecessary positive predictions.

These findings show that transfer learning is effective for this task, but not in a uniform way. For ResNet50, transfer learning produced clear gains over scratch training across accuracy, recall, F1-score, and AUC. For DenseNet121, however, the scratch model slightly outperformed the transfer-learning model in the main test metrics, while the transfer-learning model became more conservative and achieved higher precision and specificity. This reinforces the thesis claim that architecture choice and training strategy should be evaluated together rather than treated as independent design choices.

5.2 Implications for Medical Image Classification

The results have several implications for practical chest X-ray classification research at the capstone level. First, they show that a relatively focused binary classification problem can still surface meaningful tradeoffs between model architectures. Even when two models achieve similar AUC values, their recall, precision, and specificity can differ substantially. This is important because medical imaging tasks are rarely judged by accuracy alone. In a clinical screening setting, a model that misses too many abnormal images may be less useful even if its overall accuracy remains acceptable.

Second, the Grad-CAM results provide moderate evidence that the selected best-balance model learned clinically plausible spatial patterns. In the correctly classified cases, the heatmaps concentrated mainly on thoracic regions instead of obvious background areas, which supports the claim that the model relied on image content relevant to the chest. This improves the interpretability of the result and makes the model behavior easier to explain in the thesis and in the defense. However, the false positive and false negative examples also demonstrate that plausible visual attention does not guarantee correct reasoning. Grad-CAM should therefore be treated as a transparency tool rather than as proof that the model has learned clinically reliable causal structure.

Third, the completed experiments reinforce the value of keeping the project scope con-

trolled. By restricting the task to chest X-ray binary classification with two architectures and one explainability method, the project was able to produce complete training results, interpretable visual evidence, and a clear comparison that can be defended academically. This is preferable to attempting a much broader medical imaging project that would risk incomplete implementation or weak experimental depth.

5.3 Limitations

Several limitations should be acknowledged when interpreting the final results. The first is dataset quality. NIH ChestXray14 is a widely used public benchmark, but its labels were derived automatically from radiology reports rather than assigned manually for this exact classification task. As a result, label noise is likely present. This limits the confidence with which the measured performance can be interpreted as a true estimate of clinical diagnostic quality.

The second limitation is the binary task formulation itself. In this thesis, all non-“No Finding” labels were grouped into a single abnormal class. This simplification was useful for making the capstone achievable and for supporting a clean comparison across models, but it also removes important clinical detail. Different abnormalities have different visual signatures and different levels of difficulty. A model that performs reasonably on binary normal-versus-abnormal classification may still struggle on more realistic multi-label or disease-specific tasks.

The third limitation is experimental coverage. Although the full four-way comparison was completed, the project still examines only two model architectures and one binary task formulation. The conclusions therefore should not be generalized to all deep learning approaches for chest X-ray analysis. Additional architectures, alternative thresholds, and disease-specific tasks could reveal different performance patterns.

The fourth limitation concerns explainability. Grad-CAM provides an intuitive visualization of the image regions that most influence the prediction, but it remains a qualitative method. Heatmaps can look plausible even when the underlying prediction is wrong, and different explainability methods may highlight different regions. For this reason, the Grad-CAM analysis in this thesis should be interpreted as supportive evidence about model attention rather than as a definitive account of how the model reasons.

5.4 Conclusion

This thesis investigated whether transfer learning and explainable deep learning could improve chest X-ray diagnostic classification in a focused capstone setting. The study used the NIH ChestXray14 dataset and formulated the task as binary classification, where images labeled “No Finding” were treated as normal and images with any pathology label were treated as abnormal. Two convolutional neural network architectures, ResNet50 and DenseNet121, were selected as the main comparison models because both are well established in image classification and have been widely used in medical imaging research. Four controlled experiments were completed in total: ResNet50 from scratch, ResNet50 with transfer learning, DenseNet121 from scratch, and DenseNet121 with transfer learning.

The final experimental results show that the best outcome depends on how performance is defined. DenseNet121 trained from scratch achieved the highest test AUC at 0.7457, while ResNet50 with transfer learning delivered the strongest overall balance of accuracy, recall, and F1-score, reaching a test accuracy of 0.6955, F1-score of 0.6547, and AUC of 0.7454. DenseNet121 with transfer learning produced a similar AUC of 0.7421 and the highest precision and specificity, but at the cost of noticeably lower recall. ResNet50 from scratch served as the weakest of the four runs, with a test AUC of 0.7225. These results indicate that the four configurations reached different tradeoff points under the same binary task formulation, and that transfer learning was clearly beneficial for ResNet50 but not

uniformly beneficial for DenseNet121.

The Grad-CAM analysis further supported the practical value of the selected best-balance model. In representative correct predictions, the heatmaps concentrated primarily on thoracic regions rather than on irrelevant image background, suggesting that the model learned clinically plausible visual patterns. At the same time, the false positive and false negative cases showed that plausible attention maps do not guarantee correct decisions. This confirms that explainability should be used to support interpretation of model behavior rather than to replace quantitative evaluation.

Overall, the main implication of this thesis is that a carefully scoped workflow combining controlled experimentation, reproducible evaluation, and explainability can produce academically meaningful results for medical image analysis at the capstone level. Even with a simplified binary problem and only two model architectures, the study was able to reveal clear tradeoffs between sensitivity, specificity, precision, architecture choice, and training strategy. Most importantly, the results show that transfer learning improved ResNet50 substantially, but its benefit was not uniform across architectures. Grad-CAM also improved interpretability by helping connect quantitative model behavior with representative correct and incorrect predictions.

5.5 Future Work

Several directions can extend this work in a more advanced study. First, the binary classification setting can be expanded into multi-label classification so that individual thoracic diseases are modeled directly instead of being merged into a single abnormal class. This would make the task more clinically realistic and allow model performance to be analyzed at the disease level.

Second, the experimental comparison can be broadened. Additional architectures such as

EfficientNet, vision transformers, or chest X-ray models trained with self-supervised learning could be evaluated to determine whether the observed ResNet50–DenseNet121 tradeoff generalizes to more recent model families. Future work should also test separate learning-rate schedules for scratch training and transfer-learning fine-tuning, since pretrained models may benefit from a more conservative fine-tuning rate than randomly initialized models. Threshold optimization and calibration should also be explored so that the present accuracy-recall-specificity tradeoffs can be tuned more explicitly for screening or decision-support use cases.

Third, explainability can be improved beyond the present Grad-CAM analysis. Future work could compare Grad-CAM with other explainability methods, incorporate uncertainty estimation, or involve expert review of model heatmaps. This would help determine whether the highlighted regions are not only visually plausible, but also clinically meaningful from a radiological perspective.

Finally, the generalizability of the findings should be tested on additional datasets and settings. External validation on benchmarks such as CheXpert or other institutional chest X-ray collections would provide stronger evidence about robustness across domains. If the long-term goal is clinical deployment or decision support, future work should also examine threshold selection, calibration, domain shift, and user-facing evaluation rather than relying on a single fixed test setting.

Chapter A

Supplementary Materials

A.1 Training Configuration Summary

Table [A.1](#) summarizes the shared training configuration used across the four main experiments. These settings are reported here for convenience so that the main methodology chapter can remain focused on the core design choices.

Table A.1: Shared training configuration for the four main experiments.

Item	Setting
Dataset	NIH ChestXray14
Task	Binary classification (Normal vs. Abnormal)
Input size	224×224
Batch size	16
Optimizer	Adam
Loss function	BCEWithLogitsLoss
Maximum epochs	20
Early stopping patience	5
Validation criterion	AUC
Main architectures	ResNet50 and DenseNet121
Training modes	From scratch and transfer learning

A.2 Stored Experiment Artifacts

The project repository and local experiment folders retain several artifacts that support reproducibility. These include per-epoch training histories, experiment summaries, saved checkpoints, confusion matrices, ROC curves, and Grad-CAM visualizations. The materials were used to populate the result tables and qualitative analysis presented in Chapters 4 and 5.

- Training histories were saved as CSV files for each completed run.
- Best and latest checkpoints were stored separately to support evaluation and resumed training.
- Figure outputs included training curves, confusion matrices, ROC curves, and Grad-

CAM overlays.

A.3 Additional Note on Grad-CAM Examples

The main text includes representative Grad-CAM figures for correct normal, correct abnormal, false positive, and false negative predictions. Additional visual examples were generated during analysis, but only a compact subset was included in Chapter 4 in order to keep the discussion focused. The selected examples were chosen to illustrate both clinically plausible attention patterns and the limitations of relying on heatmaps alone when interpreting model behavior.

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